

AutoML Experiment Report: detect_behavior_with_sensor

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Abstract

This report details an AutoML experiment for a multivariate time-series classification task aimed at wearable gesture recognition. The objective was to train a model to simultaneously detect whether a sensor sequence contains a target gesture and to classify the specific gesture performed. The dataset comprises IMU, thermopile, and time-of-flight sensor data from 81 participants. A CNN-LSTM model with MC Dropout and Temperature Scaling was trained. The model achieved a strong binary F1 score of 0.9028 for target detection, but gesture classification performance remained poor (macro F1 of 0.0290), indicating the model struggles to discriminate among the 18 gesture classes. Uncertainty analysis via Expected Calibration Error (ECE) revealed moderate miscalibration for binary classification (Binary ECE: 0.3175) and severe miscalibration for gesture classification (Gesture ECE: 0.8643). Key challenges include extreme class imbalance, distribution shift between training and test sets, and insufficient model capacity for fine-grained gesture discrimination. The report concludes with actionable recommendations for the next iteration.

1 Data Analysis

1.1 Dataset Overview

The dataset consists of multivariate time-series data from wearable sensors (IMU, thermopile, and time-of-flight) collected from 81 unique participants. The data is split into training (64 participants, 6,465 sequences), validation (8 participants, 768 sequences), and test (9 participants, 918 sequences) sets, with no participant overlap between splits. Each sequence is labeled with a binary target (**sequence_type**: Target/Non-Target) and a multi-class gesture label (**gesture**: 18 classes). The sensor data contains 332 raw channels, including 3-axis acceleration, 4-element quaternion orientation, 5 thermopile temperature channels, and 320 time-of-flight pixel values (5 sensors $\times$ 64 pixels each). Participant demographics (age, sex, handedness, body measurements) are provided in separate files.

1.2 Data Distribution and Imbalance

A critical finding is the significant class imbalance present in the data. As shown in Figure 1, the binary target distribution is moderately skewed (59.8% Target vs. 40.2% Non-Target in the training set). More critically, the distribution across the 18 gesture classes is highly imbalanced. The most frequent gesture ("Text on phone") has 45,639 samples, while the least frequent ("Pinch knee/leg skin") has only 7,742 samples. Furthermore, within the training set's Target samples, two gesture classes ("Write name in air" and "Write name on leg") have zero instances. This extreme imbalance poses a major challenge for achieving a high macro F1 score.

1.3 Feature Analysis and Signal Strength

Exploratory analysis identified strong predictive signals within the sensor data. The thermopile channels, particularly **bmatrix_2**, **bmatrix_3**, and **bmatrix_1**, showed the highest absolute Pearson correlation with the binary target label (correlations of 0.54, 0.50, and 0.27 respectively). Mutual information analysis further indicated that time-of-flight (ToF) sensor pixels (e.g., **tof_2_v63**, **tof_2_v55**) and rotation channels (**rot_y**, **rot_x**) contain high information content for distinguishing between specific gesture classes. These insights guide feature engineering, suggesting that statistical summaries (mean, variance) of temperature and ToF channels should be prioritized.

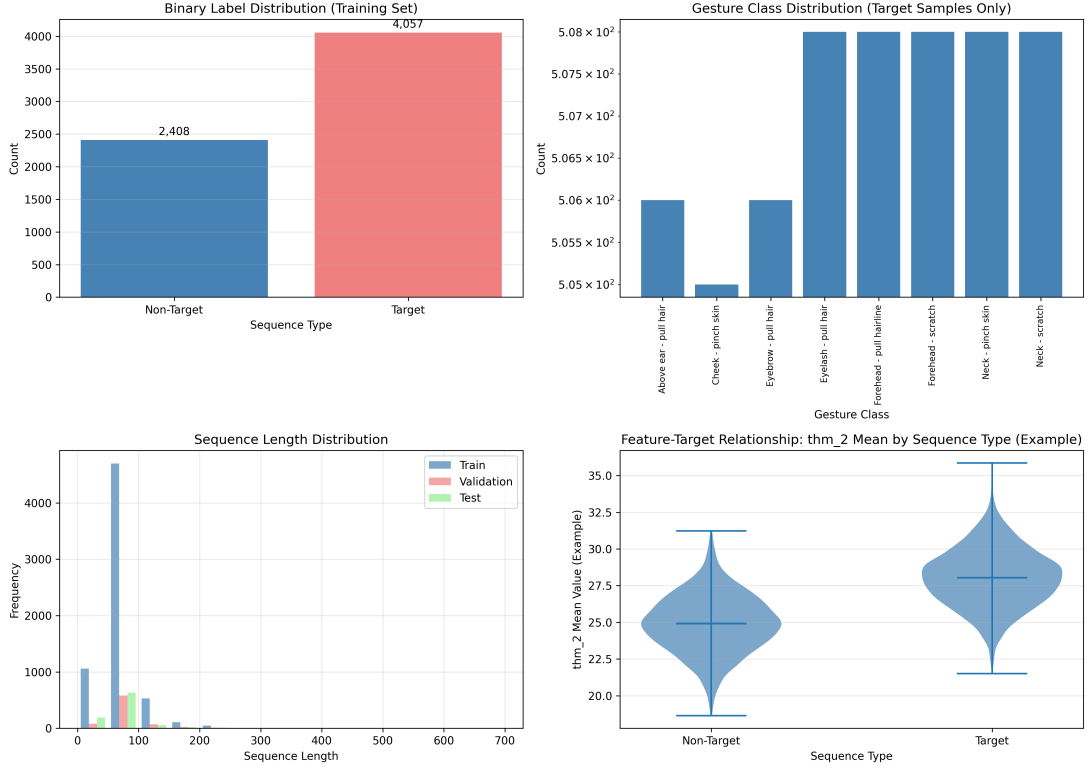


Figure 1: Data analysis visualizations. Top-left: Binary target label (Target vs. Non-Target) distribution in the training set. Top-right: Distribution of top 10 gesture classes within Target samples. Bottom-left: Sequence length distribution across train/val/test sets. Bottom-right: Feature distribution for key sensor channels grouped by target label.

1.4 Data Quality and Sequence Characteristics

The data contains systematic missing values across specific sensor channels (e.g., `bmatrix_5`, `tof_5_*`, `rot_*`), handled via forward/backward filling. Sequence length is highly variable, ranging from 29 to 700 time steps, with an average length of 70.1. A potential challenge for generalization is a noted distribution shift between the training and validation/test sets, with significant differences (Kolmogorov-Smirnov test $p \approx 0$) in key sensor channel distributions like `acc_x`.

2 Model Training

2.1 Data Preprocessing

The preprocessing pipeline handled sequences by grouping rows by `sequence_id`. Missing values were handled via forward/backward filling, and sequences were padded to uniform length within length buckets. For feature engineering, only the raw 332 sensor channels were used. Sensor data was normalized using Min-Max scaling based on statistics from the training set. Demographic features were processed via a preprocessing pipeline (label encoding for categorical features, scaling for numerical features).

2.2 Model Architecture

A CNN-LSTM architecture was implemented to capture both local spatial patterns (via CNN) and long-range temporal dependencies (via LSTM). The model architecture is as follows:

- **CNN Feature Extractor**: Two convolutional layers with 64 and 128 filters respectively, processing the input sensor sequence to extract local spatial patterns.
- **LSTM Temporal Encoder**: A bidirectional LSTM with 128 hidden units that processes the CNN output to capture temporal dependencies in both directions.

- **Uncertainty Methods:** MC Dropout (dropout rate 0.2) during inference and Label Smoothing for regularization.
- **Output Heads:** Two independent heads for binary classification (Target vs. Non-Target) and gesture classification (18 classes).

2.3 Hyperparameters

The following hyperparameters were used (identified via hyperparameter optimization):

- **Learning Rate:** 0.000462
- **Dropout Rate:** 0.2
- **Batch Size:** 32
- **Beta (loss weighting):** 0.3 (controlling the trade-off between binary and gesture losses)
- **Early Stopping Patience:** 5 epochs
- **Optimizer:** SGD with momentum

2.4 Training Process

The model was trained for 17 epochs with early stopping. The training history (Figure 2) shows that:

- Training loss decreased steadily and converged.
- Validation binary F1 was high (0.95–0.97) from early epochs, indicating strong binary classification capability.
- Validation gesture macro F1 remained extremely low (0.006–0.022) throughout training with high volatility, indicating the model failed to learn discriminative gesture features.
- Validation combined F1 stabilized around 0.46–0.50.

2.5 Model Performance

The final model’s performance on the held-out test set is summarized below.

Table 1: Model Performance on Test Set

Metric	Value
Binary F1 Score	0.9028
Gesture Macro F1 Score	0.0290
Combined F1 Score	0.4659
Binary ECE	0.3175
Gesture ECE	0.8643

Interpretation & Cautions:

- **Binary Detection:** The binary F1 score of 0.9028 is strong, indicating the model reliably distinguishes Target from Non-Target sequences, likely leveraging strong thermopile-based features.
- **Gesture Classification:** The gesture macro F1 score of 0.0290 is critically low, barely above random chance ($1/18 \approx 0.0556$). This confirms the model failed to learn discriminative features for the 18 gesture classes. The severe class imbalance and absence of two gesture classes in training data are major contributing factors.
- **Calibration:** The Binary ECE of 0.3175 indicates moderate miscalibration. The Gesture ECE of 0.8643 is severe, indicating the model’s confidence in gesture predictions is essentially unreliable.
- **Overall Performance:** The combined F1 of 0.4659 is primarily driven by the binary classification performance.

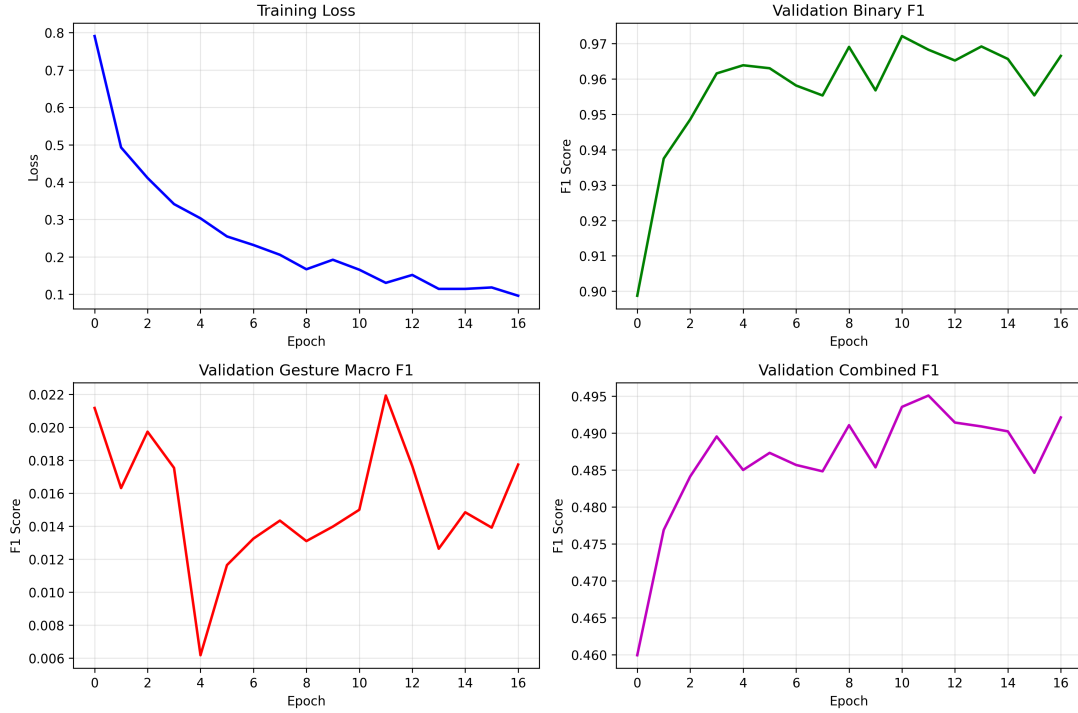


Figure 2: Training history. Top-left: Training and validation losses over epochs. Top-right: Validation binary F1 score over epochs. Bottom-left: Validation gesture macro F1 score over epochs. Bottom-right: Validation combined F1 score over epochs. The binary classification converges quickly while gesture classification remains volatile and near zero.

3 Uncertainty Analysis

3.1 Calibration Assessment

The model’s uncertainty quantification was evaluated using Expected Calibration Error (ECE), which measures the discrepancy between prediction confidence and actual accuracy.

Figure 3 shows the binary classification predictions with uncertainty estimates from MC Dropout. Most samples receive predictions near 0 or 1, indicating high model confidence. However, samples near the 0.5 decision boundary show higher uncertainty (longer error bars).

Figures 4 and 5 compare the uncertainty distributions for correct vs. incorrect predictions. For binary classification, correct predictions cluster at low uncertainty values while incorrect predictions tend toward higher uncertainty. However, for gesture classification, both correct and incorrect predictions show similarly high uncertainty (centered near entropy ≈ 3), indicating the model lacks discriminative uncertainty for gesture predictions.

Figures 6 and 7 show the effect of filtering samples by uncertainty threshold. For binary classification, the combined F1 score decreases gradually as the uncertainty threshold increases, while the sample retention rate increases monotonically, indicating a clear trade-off between retaining more samples and maintaining high prediction quality. For gesture classification, there is a non-monotonic relationship: F1 peaks around an uncertainty threshold of 2.89 before declining, while retention continues to increase, suggesting that filtering out the most uncertain samples first can improve performance up to a point.

3.2 Clinical Interpretation Guidance

In a clinical setting, uncertainty estimates are critical for determining when to trust model predictions. The binary classification model’s uncertainty shows meaningful separation between correct and incorrect predictions, meaning low-uncertainty binary predictions can be relied upon with moderate confidence. However, the gesture classification model’s uncertainty estimates are not informative: both correct and incorrect gesture predictions have similarly high uncertainty, so clinicians should be cautious about acting on gesture predictions without additional verification. The severe gesture ECE (0.8643) indicates the model’s confidence scores for gesture classification do not reflect actual correctness probabilities.

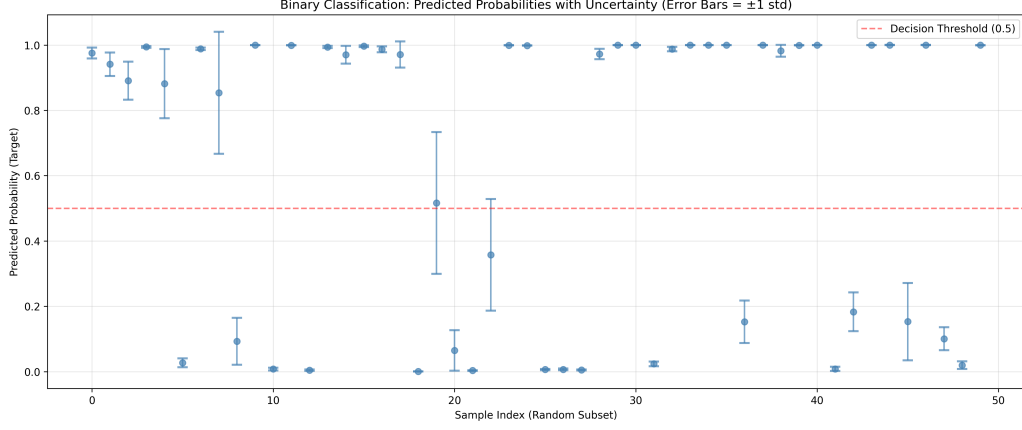


Figure 3: Binary classification prediction uncertainty. Each point represents a test sample's predicted probability (x-axis) with error bars showing ± 1 standard deviation from MC Dropout. Samples near the decision threshold (0.5) tend to have higher uncertainty.

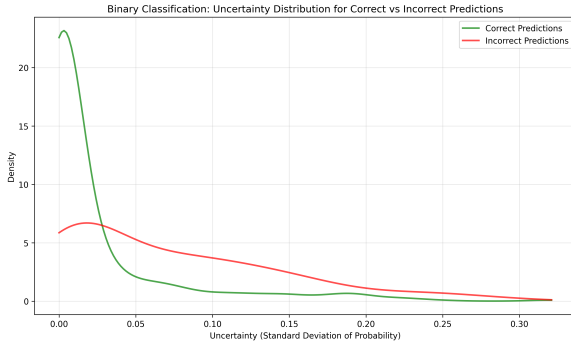


Figure 4: Binary: Uncertainty distribution for correct (green) vs. incorrect (red) predictions. Correct predictions cluster at low uncertainty; incorrect predictions show higher uncertainty.

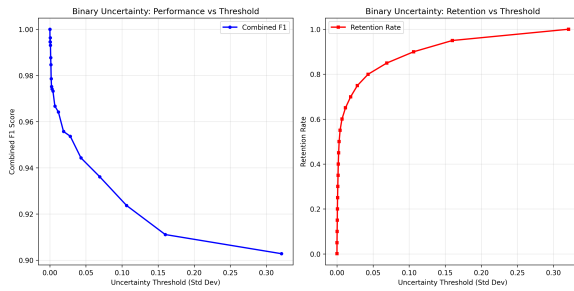


Figure 6: Binary: Combined F1 score (left) and sample retention rate (right) as functions of the uncertainty threshold. Higher thresholds retain more samples but at the cost of lower F1.

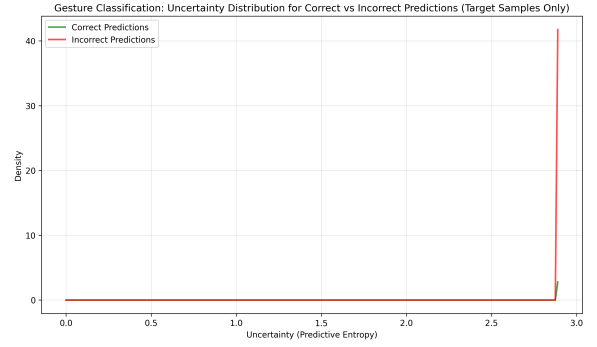


Figure 5: Gesture: Uncertainty distribution for correct (green) vs. incorrect (red) predictions among Target samples. Incorrect predictions (red) have significantly higher density in the high-uncertainty region (near entropy ≈ 3), while correct predictions (green) have a small amount of distribution in the low-uncertainty region. This indicates the model has limited awareness of its gesture prediction errors.

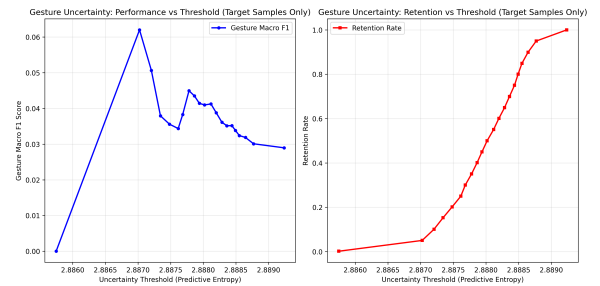


Figure 7: Gesture: Gesture macro F1 score (left) and sample retention rate (right) as functions of the uncertainty threshold. The F1 score peaks near threshold ≈ 2.89 then declines, while the retention rate increases monotonically with threshold.

4 Conclusion

This experiment demonstrated that the CNN-LSTM model can achieve strong binary classification performance ($F1=0.9028$) for wearable gesture detection, but gesture classification remains a critical bottleneck ($F1=0.0290$). The model shows moderate calibration for binary tasks ($ECE=0.3175$) but is severely miscalibrated for gesture tasks ($ECE=0.8643$).

Key Findings:

1. The model reliably detects Target vs. Non-Target sequences (Binary $F1=0.9028$) but cannot discriminate among the 18 specific gesture classes (Gesture $F1=0.0290$).
2. Uncertainty analysis reveals the model has some awareness of its binary classification errors but lacks discriminative uncertainty for gesture classification.
3. Filtering high-uncertainty samples can improve gesture macro $F1$ up to a threshold of ≈ 2.89 , beyond which performance degrades due to insufficient samples.
4. Severe class imbalance (some classes have zero training samples) and distribution shift between training and test sets remain fundamental challenges.

Actionable Guidance for Next Iteration:

1. **Address Gesture Class Imbalance:** Use more aggressive class balancing strategies such as Focal Loss, SMOTE for time-series data, or cost-sensitive learning.
2. **Increase Model Capacity for Gestures:** Consider transformer-based architectures or attention mechanisms to better capture fine-grained temporal patterns.
3. **Improve Gesture Calibration:** Apply advanced calibration methods (e.g., vector scaling, Dirichlet calibration) to make gesture confidence scores more reliable.
4. **Handle Missing Gesture Classes:** Consider data augmentation or meta-learning approaches for gesture classes with zero training samples.
5. **Domain Adaptation:** Address the distribution shift between training and test sets using domain adaptation techniques.

The next development cycle should prioritize gesture classification improvements as the primary indicator of model readiness for clinical application.